

Objective:

Students will select or propose a real or realistic project and apply end-to-end project management processes, from initiation to cost estimation (topics covered in the classes), using the methodologies, models, and analytical tools taught in the course.

This assignment will simulate a Project Manager's role in planning, evaluating, and documenting all critical phases of project management.

Assignment Components (Apply Your Course Topics)

Students must produce a **comprehensive Project Management Report** covering the following sections:

1. Project Introduction

- Project title and description, vision, and objectives
- Stakeholder identification and analysis
- Perform a **SWOT analysis** for the project
- Develop a **Project Charter** including project purpose, objectives, and summary milestones

2. Project Evaluation

Perform quantitative and qualitative evaluation using the following tools:

- Net Present Value (NPV) analysis
- Return on Investment (ROI) calculation
- Weighted Scoring Model for project selection or prioritization

3. Project Scope Management

- Define and write the **Project Scope Statement**
- Identify and list key **requirements**
- Create a **Work Breakdown Structure (WBS)** (can be a simple diagram or table form)

4. Time Management

- List all project activities with durations and dependencies
- Develop a **Gantt Chart** for the full project schedule
- Identify the **Critical Path** using **Critical Path Method (CPM)**
- Perform **PERT analysis** to estimate expected project completion time

5. Cost Estimation

- Estimate total project cost using:
 - **COCOMO Model** (*for software projects*), or
 - **Function Point Analysis (FPA)** (*for system development*)
- Provide cost summary table

Deliverables

Each group must submit:

1. **Project Management Report (15–20 pages)** including:
 - All sections above
 - Clear headings, tables, and figures
2. **Supporting Charts/Tools:**
 - Gantt chart (made in Excel, MS Project, or GanttProject)
 - WBS diagram
 - CPM/PERT network diagram
 - NPV/ROI calculation sheets
 - Cost estimation table (COCOMO/FPA)
3. **Presentation (optional but recommended)** — a 10-minute summary presentation highlighting the key findings.